Fiscal Year:	FY 2013	**Task Last Updated:**	FY 01/31/2014
PI Name:	Narayanan, Vinod Ph.D.		
Project Title:	Boiling Heat Transfer Mechanism During Thermally Actuated Pumping by Asymmetric Surface Structures		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Fluid physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Corvallis	**State:**	OR
Zip Code:	97331	**Congressional District:**	5
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2008 Fluid Physics NNH08ZTT002N
Start Date:	06/01/2009	**End Date:**	09/30/2013
No. of Post Docs:	0	**No. of PhD Degrees:**	1
No. of PhD Candidates:	0	**No. of Master' Degrees:**	2
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA GRC
Contact Monitor:	Chao, David	**Contact Phone:**	216-433-8320
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Flight Program:			
Flight Assignment:	NOTE: Extended to 9/30/2013, per NSSC information (Ed., 12/18/2012) NOTE: Extension to 9/30/2013 is Pending, per NSSC and PI (Ed., 11/1/2012) NOTE: End date is 9/30/2012 per NSSC information (Ed., 2/28/2012)		
Key Personnel Changes/Previous PI:	The PIs on the project are Vinod Narayanan from Oregon State University and Sushil Bhavnani from Auburn University.		
COI Name (Institution):	Bhavnani, Sushil (Co-PI: Auburn University)		
Grant/Contract No.:	NNX09AJ98G (OSU) ; NNX09AL63G (Auburn)		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Developments in the field of electronics fabrication have led to significant miniaturization of devices. This reduction in footprint of the electronic devices coupled with increased computing power, has allowed integration of a variety of technologies in a portable package. However, these developments have also resulted in a bottleneck stemming from the need to dissipate the resulting high heat densities in the electronic packages. A particularly effective cooling technique involves immersing the electronics in a dielectric fluid with a low boiling point, like in the early supercomputers. The heat from the electronics converts the liquid to vapor providing very high heat removal rates. These rates are increased even further if the liquid is flowing. The study investigates an innovative concept wherein a heat sink with microscopic surface features in silicon is used to create a self-propelled flow of a dielectric liquid, resulting in a pump-less, compact and self-regulating flow loop. The system proposed will also be applicable to thermal management of space electronics, where power is a precious commodity. The motivation behind the reported study is to determine whether self-propulsion of fluid can be achieved in the nucleate boiling regime, where the heat transfer rates are high, using passive surface asymmetry. Hypotheses 1. Asymmetry in the forces of a growing or departed bubble can cause axial motion in a phase-change fluid 2. A net pumping effect can be realized for a surface with repeated asymmetrical structures 3. Axial motion of large vapor bubbles and slugs can be caused by asymmetric inertial force, thereby reducing the possibility of dry spots, and increasing critical heat flux limits. This is especially significant for downward facing surfaces or in microgravity environment. Objectives In order to resolve the above stated hypotheses, an experimental and analytical approach is undertaken to 1. Measure, by high-speed flow visualization, the time-varying bubble diameters and frequencies; 2. Quantify liquid velocity using particle velocimetry, and the bubble velocity; 3. Develop and validate analytical models to predict the fluid velocity.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Passive fluid motion using meso/micro- scale surface structures has been demonstrated in this project in a pool and open channel configuration. Models to predict the fluid velocity in both configurations have been developed. Self-propulsion has been observed under varying gravity conditions and for different fluids. These results will have an impact on passive thermal management in phase-change systems both in terrestrial, but more importantly, under microgravity conditions. Moreover, detailed high speed imaging observations and data analysis provides fundamental information regarding the ebullition cycle on such surfaces in the presence of highly wetting fluids. It is likely that the concept has uses beyond thermal management such as in biomedical industry for lab-on-a-chip applications and in portable chemical plants. Passive means of moving fluids is crucial to the compactness and reliability in these fields.
Task Progress:	Major Activities Performed: In this study, a silicon heat sink with an asymmetric saw-tooth cross-sectioned surface structure was fabricated using a combination of gray-scale lithography, deep reactive ion etching and wet etching techniques. The novelty of the heat sink lies in the ability to effect lateral motion of bubbles due to nucleation from re-entrant cavities fabricated on the shallow slope of the saw-toothed surface. The asymmetric nucleation, growth and departure of bubbles leads to an angular momentum imparted to the liquid, thereby resulting in a net lateral flow. The study investigates the ability of surface structure to propel the liquid in its immediate vicinity under a variety of test conditions including microgravity. Both pool and open channel configurations were studied with asymmetric ratchets. Experimental and analytical work was performed both at OSU and Auburn during this project. A significant effort was put forth in preparation and execution of microgravity flight experiments in May 2012. Specific Objectives: An experimental and analytical approach was undertaken to 1. Measure, by high-speed flow visualization, the time-varying bubble diameters and frequencies 2. Quantify liquid velocity using particle velocimetry, and the bubble velocity 3. Develop and validate analytical models to predict the fluid velocity. Significant Results: 1. Passive directional fluid motion in pool and open channel configurations during boiling was observed. This directional motion was observed under different geometrical, gravitational, and fluidic parameters. 2. Models were put forth to explain the resultant motion for pool and open channel configurations and for a microgravitry environment. Due to the unique profile of the surface, the bubble characteristics are very different from those reported in the literature for common surfaces and fluids. In the experiments conducted, bubble growth and departure from re-entrant cavities on the asymmetric structures were studied using high speed photography and image processing techniques. Bubble dynamics such as growth rate, bubble departure diameter and frequency were studied as a function of heat flux and subcooling. Asymptotic growth relationships were expressed, for the inertia and heat transfer controlled regimes, respectively. To understand the effects of gravity on bubble dynamics and self-propelled bubble motion across the surface, pool boiling experiments were conducted aboard NASA's reduced gravity flight. In FC-72, vapor bubbles six times larger in diameter compared to terrestrial gravity were observed, due to lack of buoyancy. Interestingly, bubbles were observed to be sliding across the asymmetric surface at velocities as high as 27.4 mm/s. In comparison, pool boiling on a plain surface would result in stagnant, surface- residing vapor bubbles that would affect the heat transfer characteristics adversely causing burn-out of the chip being cooled. In the current study, the ability to move the bubble along the surface at high velocities prevented any heat transfer deterioration, and actually led to an enhancement. The sliding

	motion was attributed to pressure differences in the thin liquid film existing between the saw-toothed surface and the vapor bubble. A model was proposed for the estimation of vapor bubble sliding velocity. The following manuscripts are in preparation to be submitted early in 2014: 1. Strid, L., Thiagarajan, N., Narayanan, V., Bhavnani, S. H., Linke, H., 2013, “Self-propelled Slug Flow during Boiling Using Asymmetric Meso-scale Texturing,” in preparation, to be submitted to Applied Physics Letters in February 2014. 2. Kapsenberg, F., Strid, L., Thiagarajan, N., Narayanan, V., and Bhavnani, S. H., 2013, “On the Lateral Fluid Motion During Pool Boiling via Asymmetrically Located Cavities,” in preparation, to be submitted to Applied Physics Letters in February 2014. 3. Thiagarajan, N., Bhavnani, S. H., Strid, L., and Narayanan, V., 2013, “Self-propelled Bubble Motion in Pool Boiling on Asymmetric Micro-structures under Reduced Gravity,” in preparation, to be submitted to International Journal of Multiphase Flow in February 2014.
Bibliography Type:	Description: (Last Updated: 02/03/2014)
Articles in Peer-reviewed Journals	Thiagarajan N, Strid L, Narayanan V, Bhavnani SH. "Bubble Growth During Pool Boiling of FC-72 on Asymmetric Silicon Micro-structured Surfaces with Reentrant Cavities." Submitted to International Journal of Heat and Mass Transfer, October 2013. , Oct-2013
Dissertations and Theses	Thiagarajan N. "Investigation of Thermally-Actuated Pumping During Pool Boiling of a Dielectric Liquid on an Asymmetric Microstructured Silicon Heat Sink." PhD dissertation, Auburn University, Auburn, AL, May 2013. , May-2013
Dissertations and Theses	Strid L. "Passive Pumping in Pool and Open Channel Configurations via Meso-Scale Asymmetric Surface Patterning." MS Thesis, Oregon State University, Corvallis, OR, May 2013. , May-2013
Patents	Provisional Patent Application No.: 61894780. Filed on October 23, 2013. Oct-2013 Narayanan V, Strid L, Bhavnani SH, Thiagarajan N, Natesh S, Linke H. "Passive Phase Change Heat Transport Using Meso and Micro Structures."